

Comprehensive Evaluation of Quantization Methods for Retrieval-Augmented Generation Systems

1 Research Design

1.1 Research Questions

- **RQ1:** Does quantization affect RAG answer accuracy compared to parametric knowledge?
- **RQ2:** Does quantization increase hallucination rates and reduce context faithfulness?
- **RQ3:** How does quantization impact performance across varying context lengths?
- **RQ4:** Does quantization affect output consistency across multiple runs?
- **RQ5:** How does quantization affect robustness to distractor passages?

1.2 Model Configurations

Table 1: Evaluated Model Configurations

Configuration	Base Model	Quantization	Size (GB)
FP16 Base	mistralai/Mistral-7B-v0.1	None	13.49
FP16 Instruct	mistralai/Mistral-7B-Instruct-v0.1	None	13.49
AWQ Base	TheBloke/Mistral-7B-v0.1-AWQ	4-bit AWQ	3.12
AWQ Instruct	TheBloke/Mistral-7B-Instruct-v0.1-AWQ	4-bit AWQ	3.12
NF4 Base	mistralai/Mistral-7B-v0.1	4-bit NF4	3.12
NF4 Instruct	mistralai/Mistral-7B-Instruct-v0.1	4-bit NF4	3.12
GPTQ Base	TheBloke/Mistral-7B-v0.1-GPTQ	4-bit GPTQ	3.12
GPTQ Instruct	TheBloke/Mistral-7B-Instruct-v0.1-GPTQ	4-bit GPTQ	3.12

Quantization Details:

- **AWQ:** 4-bit weights, group size 128, zero-point quantization, per-group scaling
- **GPTQ:** 4-bit weights, group size 128, asymmetric quantization
- **NF4:** 4-bit NormalFloat, double quantization, on-the-fly via BitsAndBytes
- **Compression:** All quantized methods achieve $4.32\times$ compression vs FP16

Model Architecture (Mistral-7B):

- Parameters: 7.24B
- Layers: 32
- Hidden size: 4096
- Attention heads: 32 (128-dim each)
- FFN intermediate: 14,336
- Vocabulary: 32,000 (SentencePiece)
- Max sequence length: 4096 tokens

1.3 Generation Settings

Table 2: Generation Hyperparameters

Parameter	Value
max_new_tokens	128
do_sample	True
temperature	0.7 (instruct), 0.3 (base)
top_k	50
top_p	0.9
repetition_penalty	1.1 (instruct), 1.3 (base)
random_seed	42

1.4 Evaluation Sets Overview

Table 3: Evaluation Sets Summary

Set	Samples	Gens/Model	Purpose
Set A	100	200	Answer quality & faithfulness
Set B	240	240	Context length robustness
Set C	30	150	Output consistency
Set D	75	150	Distractor sensitivity
Total	445	740	

Total generations: 740 per model $\times$ 8 configs = 5,920

2 Prompt Format Selection Study

2.1 Motivation

Prior to the main RAG evaluation, we conducted a systematic study to identify optimal prompt formats for each model variant. Prompt formatting significantly impacts model performance, particularly for quantized models where degraded instruction-following capabilities may interact with prompt structure. This preliminary study ensures fair comparison across quantization methods by selecting the best-performing format for each configuration.

2.2 Design

Purpose: Identify the optimal prompt format for base and instruction-tuned model variants across all quantization methods.

Evaluated Formats:

Instruction-tuned models (5 formats):

- **Minimal:** [INST] {context}\n\n{question} [/INST]
- **Verbose:** Explicit instructions with "Answer:" label
- **Chat:** [INST] Context: {context}\n\nBased on this, {question} [/INST]
- **System:** Includes «SYS» block with role description
- **QA:** [INST] Context: {context}\n\nQ: {question}\nA: [/INST]

Base models (5 formats):

- **Minimal:** {context}\n\nQ: {question}\nA:
- **Few-shot (1 example):** Single demonstration before target question
- **Few-shot (2 examples):** Two demonstrations before target question
- **Labeled:** Context: {context}\n\nQuestion: {question}\nAnswer:
- **Completion:** {context}\n\nThe answer to "{question}" is:

Test Set:

- 30 questions stratified across 6 categories (5 per category)
- Categories: simple factual, multi-hop, numerical, context-dependent, unanswerable, long-form
- Sources: SQuAD v2 (validation), HotpotQA (validation), DROP (validation), NarrativeQA (validation), ELI5 (validation)
- **Total:** 8 configs $\times$ 5 formats $\times$ 30 questions = 1,200 generations

Evaluation Metrics:

- **Pass/Fail Binary:** Answer correctness + output quality checks
- Quality checks: token loops, excessive special characters, reasonable length, lexical diversity
- Answer matching: fuzzy string matching (rapidfuzz partial_ratio ≥ 0.85) + exact substring matching

2.3 Results

2.3.1 Overall Format Performance

Table 4: Pass Rate by Format and Model Configuration

Configuration	Minimal	Format 2	Format 3	Format 4	Format 5
FP16 Base	0.44	0.28	0.44	0.48	0.48
FP16 Instruct	0.48	0.56	0.60	0.52	0.40
AWQ Base	0.40	0.36	0.44	0.44	0.40
AWQ Instruct	0.48	0.52	0.40	0.56	0.44
NF4 Base	0.48	0.32	0.48	0.48	0.44
NF4 Instruct	0.40	0.48	0.52	0.52	0.44
GPTQ Base	0.36	0.04	0.00	0.28	0.32
GPTQ Instruct	0.16	0.00	0.40	0.00	0.40

Base Format 2=Few-shot(1ex), 3=Few-shot(2ex), 4=Labeled, 5=Completion

Instruct Format 2=Verbose, 3=Chat, 4=System, 5=QA

Table 5: Performance Variance Across Formats

Configuration	Best	Worst	Range	Interpretation
FP16 Base	0.48	0.28	0.20	Moderate sensitivity
FP16 Instruct	0.60	0.40	0.20	Moderate sensitivity
AWQ Base	0.44	0.36	0.08	Low sensitivity
AWQ Instruct	0.56	0.40	0.16	Moderate sensitivity
NF4 Base	0.48	0.32	0.16	Moderate sensitivity
NF4 Instruct	0.52	0.40	0.12	Low sensitivity
GPTQ Base	0.36	0.00	0.36	High sensitivity
GPTQ Instruct	0.40	0.00	0.40	High sensitivity

Table 6: Category-Specific Performance by Format (FP16 Base)

Category	Minimal	Few-1	Few-2	Labeled	Compl
Simple Factual	0.68	0.20	0.60	0.68	0.68
Multi-hop	0.44	0.20	0.44	0.44	0.44
Numerical	0.44	0.60	0.40	0.44	0.44
Context-dep	0.52	0.40	0.52	0.52	0.52
Unanswerable	0.04	0.00	0.04	0.04	0.00

Table 7: Category-Specific Performance by Format (FP16 Instruct)

Category	Minimal	Verbose	Chat	System	QA
Simple Factual	0.88	0.88	0.88	0.88	0.88
Multi-hop	0.56	0.56	0.56	0.56	0.56
Numerical	0.52	0.44	0.52	0.52	0.52
Context-dep	0.40	0.48	0.40	0.40	0.40
Unanswerable	0.00	0.20	0.20	0.20	0.00

Table 8: Category-Specific Performance by Format (AWQ Base)

Category	Minimal	Few-1	Few-2	Labeled	Compl
Simple Factual	0.64	0.36	0.64	0.64	0.64
Multi-hop	0.40	0.40	0.40	0.40	0.40
Numerical	0.44	0.36	0.44	0.44	0.44
Context-dep	0.52	0.52	0.52	0.52	0.52
Unanswerable	0.04	0.00	0.00	0.04	0.00

Table 9: Category-Specific Performance by Format (AWQ Instruct)

Category	Minimal	Verbose	Chat	System	QA
Simple Factual	0.92	0.92	0.92	0.92	0.92
Multi-hop	0.48	0.48	0.32	0.48	0.48
Numerical	0.44	0.52	0.12	0.44	0.44
Context-dep	0.32	0.32	0.32	0.32	0.32
Unanswerable	0.24	0.24	0.32	0.24	0.00

Table 10: Category-Specific Performance by Format (NF4 Base)

Category	Minimal	Few-1	Few-2	Labeled	Compl
Simple Factual	0.60	0.20	0.60	0.60	0.60
Multi-hop	0.64	0.64	0.64	0.64	0.64
Numerical	0.40	0.00	0.40	0.40	0.20
Context-dep	0.52	0.40	0.52	0.52	0.52
Unanswerable	0.04	0.00	0.04	0.04	0.04

Table 11: Category-Specific Performance by Format (NF4 Instruct)

Category	Minimal	Verbose	Chat	System	QA
Simple Factual	1.00	1.00	1.00	1.00	1.00
Multi-hop	0.48	0.48	0.48	0.48	0.48
Numerical	0.40	0.36	0.40	0.40	0.40
Context-dep	0.24	0.24	0.24	0.24	0.24
Unanswerable	0.24	0.24	0.24	0.24	0.00

Table 12: Category-Specific Performance by Format (GPTQ Base)

Category	Minimal	Few-1	Few-2	Labeled	Compl
Simple Factual	0.40	0.00	0.00	0.40	0.40
Multi-hop	0.12	0.00	0.00	0.12	0.12
Numerical	0.28	0.00	0.00	0.28	0.00
Context-dep	0.12	0.12	0.00	0.12	0.12
Unanswerable	0.08	0.08	0.00	0.08	0.00

Table 13: Category-Specific Performance by Format (GPTQ Instruct)

Category	Minimal	Verbose	Chat	System	QA
Simple Factual	0.48	0.00	0.48	0.00	0.48
Multi-hop	0.00	0.00	0.00	0.00	0.00
Numerical	0.12	0.00	0.12	0.00	0.12
Context-dep	0.08	0.00	0.08	0.00	0.08
Unanswerable	0.28	0.00	0.28	0.00	0.28

3 Set A: Answer Quality & Faithfulness

3.1 Design

Purpose: Measure accuracy and hallucination rates (RQ1, RQ2)

Sample Composition:

- Factual questions: 30 (SQuAD v2)
- Multihop questions: 25 (HotpotQA)
- Numerical questions: 20 (DROP)
- Unanswerable questions: 25 (SQuAD v2)
- **Total:** 100 samples

Conditions per sample:

- RAG: Question + context (1024 tokens max)
- No-RAG: Question only (parametric knowledge)
- **Total generations:** 100 samples $\times$ 2 conditions = 200 per model

Prompt Templates:

Base Models - RAG:

{context}\n\nQ: {question}\nA:

Base Models - No-RAG:

Q: {question}\nA:

Instruction Models - RAG:

[INST] {context}\n\n{question} [/INST]

Instruction Models - No-RAG:

[INST] {question} [/INST]

3.2 Metrics

Accuracy Metrics:

- **Exact Match (EM):** Binary string equality after normalization
- **Token F1:** Token-level precision/recall, max over ground truth variants
- **Substring Match:** Whether ground truth appears in prediction
- **Token Precision:** $\frac{|\text{pred} \cap \text{ref}|}{|\text{pred}|}$
- **Token Recall:** $\frac{|\text{pred} \cap \text{ref}|}{|\text{ref}|}$
- **Jaccard Similarity:** $\frac{|\text{pred} \cap \text{ref}|}{|\text{pred} \cup \text{ref}|}$

Normalization Protocol (SQuAD-style):

```
def normalize_answer(text):
    text = unicodedata.normalize('NFD', text).lower()
    text = re.sub(r'\b(a|an|the)\b', ' ', text)
    text = re.sub(r'[^a-z0-9\s]', '', text)
    return ' '.join(text.split()).strip()
```

Extraction Strategies:

- **Raw:** Minimal processing (prompt removal only)
- **Conservative:** First line + length-based truncation ($3 \times$ ground truth)
- **Aggressive:** First sentence + first line

Faithfulness Metrics:

- **Token Overlap:** $\frac{|\text{pred tokens} \cap \text{context tokens}|}{|\text{pred tokens}|}$
- **Hallucination Score:** $1 - \text{Token Overlap}$
- **Bigram Attribution:** Fraction of prediction bigrams in context
- **Trigram Attribution:** Fraction of prediction trigrams in context
- **Context Sufficiency:** Same as token overlap (simple heuristic)

Statistical Analysis:

- Paired t-tests (RAG vs No-RAG)
- Bootstrap 95% CI (1000 iterations)
- Cohen's d effect sizes
- Bonferroni correction: $\alpha = 0.05/8 = 0.00625$

3.3 Results

3.3.1 Answer Quality: RAG vs No-RAG

Table 14: Answer Quality: RAG vs No-RAG (Conservative Extraction)

Model	RAG F1	No-RAG F1	Δ F1	$\Delta\%$	p-value	Cohen’s d
FP16 Base	0.134	0.104	+0.031	+30.5%	0.310	0.120
FP16 Instruct	0.164	0.054	+0.110	+204.3%	< 0.001***	0.530
AWQ Base	0.135	0.106	+0.029	+27.2%	0.388	0.107
AWQ Instruct	0.168	0.056	+0.112	+199.8%	< 0.001***	0.530
NF4 Base	0.135	0.104	+0.031	+29.7%	0.333	0.114
NF4 Instruct	0.157	0.054	+0.105	+199.9%	< 0.001***	0.531
GPTQ Base	0.081	0.108	-0.027	-25.2%	0.353	-0.119
GPTQ Instruct	0.099	0.056	+0.069	+224.2%	< 0.001***	0.552

*** p < 0.001
(Bonferroni-corrected)

Table 15: Exact Match Performance (Conservative Extraction)

Model	RAG EM	No-RAG EM	Δ EM	Improvement
FP16 Base	0.050	0.050	0.000	0%
FP16 Instruct	0.050	0.000	+0.050	+100%
AWQ Base	0.060	0.050	+0.010	+20%
AWQ Instruct	0.050	0.000	+0.050	+100%
NF4 Base	0.060	0.060	0.000	0%
NF4 Instruct	0.030	0.000	+0.030	+100%
GPTQ Base	0.010	0.040	-0.030	-75%
GPTQ Instruct	0.000	0.000	0.000	—

3.3.2 Per-Category Performance

Table 16: RAG Token F1 by Question Category (Conservative Extraction)

Model	Factual	Multihop	Numerical	Unanswerable
FP16 Base	0.178	0.210	0.140	0.000
FP16 Instruct	0.328	0.154	0.136	0.000
AWQ Base	0.183	0.208	0.139	0.000
AWQ Instruct	0.378	0.116	0.129	0.000
NF4 Base	0.200	0.201	0.122	0.000
NF4 Instruct	0.322	0.143	0.122	0.000
GPTQ Base	0.180	0.068	0.050	0.000
GPTQ Instruct	0.209	0.075	0.088	0.000

3.3.3 Quantization Degradation

Table 17: Quantization Degradation: Token F1 Relative to FP16 Baseline

Variant	Metric	AWQ	NF4	GPTQ
Base	Absolute Δ	+0.0006	+0.0007	-0.0531
	Relative %	-0.5%	-0.5%	+39.7%
Instruct	Absolute Δ	+0.0042	-0.0072	-0.0649
	Relative %	-2.6%	+4.4%	+39.6%

3.3.4 Faithfulness Metrics

Table 18: Faithfulness Metrics (RAG Condition)

Model	Token Overlap	Hallucination	Bigram Attr	Trigram Attr
FP16 Base	0.274	0.726	0.104	0.037
FP16 Instruct	0.618	0.382	0.332	0.178
AWQ Base	0.282	0.718	0.106	0.039
AWQ Instruct	0.604	0.396	0.299	0.157
NF4 Base	0.318	0.682	0.127	0.042
NF4 Instruct	0.583	0.417	0.283	0.140
GPTQ Base	0.300	0.700	0.101	0.058
GPTQ Instruct	0.550	0.450	0.262	0.165

3.3.5 Sample Consistency

Table 19: Sample-Level Consistency: Spearman Correlation with FP16

Variant	AWQ	NF4	GPTQ
Base	0.819	0.743	0.240
Instruct	0.957	0.973	0.845

4 Set B: Context Length Robustness

4.1 Design

Purpose: Test performance degradation across context lengths (RQ3)

Experimental Design:

- **Context Lengths:** 512, 1024, 2048, 4096 tokens
- **Answer Positions:** start, middle, end
- **Samples per config:** 20
- **Total:** 4 lengths $\times$ 3 positions $\times$ 20 samples = 240 per model

Context Padding Protocol:

```
# Base samples from SQuAD v2 (100-400 token contexts)  
# Pad to target length using WikiText-103 distractor passages  
# Place original context at specified position (start/middle/end)  
# Verify answer position post-padding (within 25% tolerance)
```

```
def pad_context(original_ctx, target_length, position):  
    distractor = sample_from_wikitext103()  
    if position == 'start':  
        return original_ctx + distractor  
    elif position == 'middle':  
        mid = len(distractor) // 2  
        return distractor[:mid] + original_ctx + distractor[mid:]  
    else: # end  
        return distractor + original_ctx
```

Prompt: RAG prompts only (same as Set A)

4.2 Metrics

Accuracy Metrics:

- Token F1, Exact Match, Substring Match
- Computed using conservative extraction

Degradation Analysis:

```
# Linear regression: F1 = beta_0 + beta_1 * length  
from scipy.stats import linregress
```

```
slope, intercept, r_value, p_value, std_err = \  
    linregress(context_lengths, f1_scores)
```

```
degradation_per_1k = slope * 1000  
relative_degradation_pct = \  
    (initial_f1 - final_f1) / initial_f1 * 100
```

Position Effects:

- Mean F1 by position (start/middle/end)
- Best/worst position identification
- Range: max - min across positions

4.3 Results

4.3.1 Performance by Context Length

Table 20: Token F1 by Context Length

Model	512	1024	2048	4096	Total Drop	Slope/1k
FP16 Base	0.213	0.399	0.403	0.210	+0.002	-0.020
FP16 Instruct	0.338	0.357	0.292	0.176	+0.162	-0.050
AWQ Base	0.229	0.473	0.387	0.264	-0.035	-0.016
AWQ Instruct	0.336	0.350	0.308	0.173	+0.162	-0.049
NF4 Base	0.195	0.371	0.430	0.205	-0.011	-0.014
NF4 Instruct	0.379	0.341	0.277	0.168	+0.211	-0.058
GPTQ Base	0.060	0.018	0.000	0.000	+0.060	-0.013
GPTQ Instruct	0.175	0.130	0.129	0.145	+0.030	-0.004

Table 21: Exact Match by Context Length

Model	512	1024	2048	4096	Total Drop	Drop %
FP16 Base	0.117	0.233	0.250	0.133	-0.017	-14.3%
FP16 Instruct	0.083	0.133	0.083	0.033	+0.050	+60.0%
AWQ Base	0.117	0.300	0.233	0.183	-0.067	-57.1%
AWQ Instruct	0.067	0.117	0.100	0.050	+0.017	+25.0%
NF4 Base	0.083	0.217	0.250	0.150	-0.067	-80.0%
NF4 Instruct	0.100	0.117	0.067	0.050	+0.050	+50.0%
GPTQ Base	0.000	0.000	0.000	0.000	0.000	0.0%
GPTQ Instruct	0.017	0.033	0.000	0.017	0.000	0.0%

Table 22: Substring Match by Context Length

Model	512	1024	2048	4096	Total Drop	Drop %
FP16 Base	0.217	0.433	0.450	0.283	-0.067	-30.8%
FP16 Instruct	0.783	0.817	0.650	0.367	+0.417	+53.2%
AWQ Base	0.283	0.483	0.367	0.233	+0.050	+17.6%
AWQ Instruct	0.833	0.800	0.617	0.383	+0.450	+54.0%
NF4 Base	0.233	0.417	0.483	0.267	-0.033	-14.3%
NF4 Instruct	0.883	0.783	0.600	0.383	+0.500	+56.6%
GPTQ Base	0.167	0.000	0.000	0.000	+0.167	+100.0%
GPTQ Instruct	0.517	0.233	0.250	0.333	+0.183	+35.5%

4.3.2 Degradation Statistics

Table 23: Context Length Degradation Analysis (Token F1)

Model	R^2	p-value	Initial	Final	Rel Drop %
FP16 Base	0.081	0.716	0.213	0.210	+1.2%
FP16 Instruct	0.945	0.028*	0.338	0.176	+47.8%
AWQ Base	0.052	0.771	0.229	0.264	-15.2%
AWQ Instruct	0.925	0.038*	0.336	0.173	+48.4%
NF4 Base	0.035	0.814	0.195	0.205	-5.4%
NF4 Instruct	0.996	0.002**	0.379	0.168	+55.7%
GPTQ Base	0.560	0.252	0.060	0.000	+100.0%
GPTQ Instruct	0.093	0.695	0.175	0.145	+17.1%

* $p < 0.05$, ** $p < 0.01$

Statistical Significance:

- **Significant Degradation:** FP16 Instruct, AWQ Instruct, NF4 Instruct (all $p < 0.05$)
- **High R^2 :** Instruction models show linear degradation (R^2 0.92-1.00)
- **Base Models:** No significant degradation ($p > 0.25$), some show improvement
- **GPTQ Base:** Complete collapse at long contexts despite moderate R^2 (0.56)

4.3.3 Answer Position Effects

Table 24: Token F1 by Answer Position (Overall)

Model	Start	Middle	End	Best Pos	Range
FP16 Base	0.243	0.306	0.369	End	0.126
FP16 Instruct	0.262	0.347	0.263	Middle	0.085
AWQ Base	0.275	0.347	0.394	End	0.119
AWQ Instruct	0.250	0.323	0.301	Middle	0.073
NF4 Base	0.225	0.292	0.384	End	0.159
NF4 Instruct	0.252	0.335	0.286	Middle	0.083
GPTQ Base	0.008	0.014	0.036	End	0.028
GPTQ Instruct	0.094	0.081	0.260	End	0.179

Table 25: Exact Match by Answer Position (Overall)

Model	Start	Middle	End	Best Pos	Range
FP16 Base	0.113	0.225	0.213	Middle	0.113
FP16 Instruct	0.063	0.113	0.075	Middle	0.050
AWQ Base	0.163	0.250	0.213	Middle	0.088
AWQ Instruct	0.050	0.088	0.113	End	0.063
NF4 Base	0.113	0.188	0.225	End	0.113
NF4 Instruct	0.063	0.113	0.075	Middle	0.050
GPTQ Base	0.000	0.000	0.000	—	0.000
GPTQ Instruct	0.013	0.000	0.038	End	0.038

4.3.4 Position Effects by Context Length

Table 26: Position Effects at 512 Tokens (Token F1)

Model	Start	Middle	End	Best
FP16 Base	0.120	0.156	0.362	End
FP16 Instruct	0.331	0.358	0.325	Middle
AWQ Base	0.169	0.193	0.326	End
AWQ Instruct	0.330	0.320	0.356	End
NF4 Base	0.084	0.150	0.350	End
NF4 Instruct	0.305	0.437	0.394	Middle
GPTQ Base	0.022	0.047	0.111	End
GPTQ Instruct	0.142	0.107	0.277	End

Table 27: Position Effects at 1024 Tokens (Token F1)

Model	Start	Middle	End	Best
FP16 Base	0.393	0.297	0.506	End
FP16 Instruct	0.325	0.419	0.327	Middle
AWQ Base	0.463	0.425	0.531	End
AWQ Instruct	0.309	0.431	0.309	Middle
NF4 Base	0.383	0.217	0.512	End
NF4 Instruct	0.319	0.435	0.269	Middle
GPTQ Base	0.010	0.010	0.034	End
GPTQ Instruct	0.030	0.062	0.300	End

Table 28: Position Effects at 2048 Tokens (Token F1)

Model	Start	Middle	End	Best
FP16 Base	0.361	0.464	0.384	Middle
FP16 Instruct	0.284	0.397	0.195	Middle
AWQ Base	0.347	0.422	0.391	Middle
AWQ Instruct	0.287	0.350	0.285	Middle
NF4 Base	0.358	0.502	0.432	Middle
NF4 Instruct	0.301	0.329	0.199	Middle
GPTQ Base	0.000	0.000	0.000	—
GPTQ Instruct	0.101	0.091	0.193	End

Table 29: Position Effects at 4096 Tokens (Token F1)

Model	Start	Middle	End	Best
FP16 Base	0.099	0.308	0.224	Middle
FP16 Instruct	0.109	0.216	0.205	Middle
AWQ Base	0.120	0.346	0.327	Middle
AWQ Instruct	0.075	0.192	0.253	End
NF4 Base	0.073	0.300	0.242	Middle
NF4 Instruct	0.084	0.139	0.280	End
GPTQ Base	0.000	0.000	0.000	—
GPTQ Instruct	0.102	0.064	0.270	End

4.3.5 Quantization Impact on Long-Context Robustness

Table 30: Quantization vs FP16: Token F1 Degradation Comparison

Variant	Method	FP16 Slope	Quant Slope	Slope Diff	Addl Degrad
Base	AWQ	-0.020	-0.016	+0.003	-0.038
	NF4	-0.020	-0.014	+0.006	-0.013
	GPTQ	-0.020	-0.013	+0.006	+0.058
Instruct	AWQ	-0.050	-0.049	+0.001	+0.001
	NF4	-0.050	-0.058	-0.008	+0.049
	GPTQ	-0.050	-0.004	+0.046	-0.132

Slope = degradation per 1000 tokens; Addl Degrad = additional total degradation vs FP16

Table 31: Quantization Impact: Accuracy at Extremes

Variant	Method	FP16@512	Quant@512	FP16@4096	Quant@4096
Base	AWQ	0.213	0.229	0.210	0.264
	NF4	0.213	0.195	0.210	0.205
	GPTQ	0.213	0.060	0.210	0.000
Instruct	AWQ	0.338	0.336	0.176	0.173
	NF4	0.338	0.379	0.176	0.168
	GPTQ	0.338	0.175	0.176	0.145

Interpretation:

- **AWQ:** Preserves long-context performance nearly identically to FP16
- **NF4:** Slightly worse degradation for instruct (+0.049), stable for base
- **GPTQ Base:** Complete failure at 2048+ tokens despite better slope
- **GPTQ Instruct:** Maintains minimal functionality but low accuracy throughout

4.3.6 Performance Ranges and Variance

Table 32: Performance Variance Across Lengths (Token F1)

Model	Min	Max	Range	Coef Var
FP16 Base	0.210	0.403	0.193	0.320
FP16 Instruct	0.176	0.357	0.181	0.283
AWQ Base	0.229	0.473	0.244	0.340
AWQ Instruct	0.173	0.350	0.177	0.290
NF4 Base	0.195	0.430	0.235	0.358
NF4 Instruct	0.168	0.379	0.211	0.318
GPTQ Base	0.000	0.060	0.060	1.635
GPTQ Instruct	0.129	0.175	0.046	0.154

Stability Analysis:

- **Lowest Variance:** GPTQ Instruct (Coef Var 0.154) - consistently poor
- **Highest Variance:** GPTQ Base (Coef Var 1.635) - catastrophic collapse
- **Instruction Models:** More stable variance (0.15-0.32) than base (0.32-1.64)
- **Peak Variability:** NF4/AWQ base models show high range but controlled variance

5 Set C: Output Consistency

5.1 Design

Purpose: Measure output variance across multiple runs with identical prompts (RQ4)

Sample Composition:

- 30 samples from Set A (stratified by category)
- 5 runs per sample (random seeds 42-46)
- Temperature: 0.7 (instruct), 0.3 (base)
- **Total:** 30 samples $\times$ 5 runs = 150 generations per model

Prompt: RAG prompts only (same as Set A)

5.2 Metrics

Consistency Metrics:

- **Self-BLEU:** Average pairwise BLEU between outputs (higher = more consistent)
- **Exact Match Consistency:** Fraction of output pairs matching exactly after normalization
- **Token Overlap:** Average Jaccard similarity across all output pairs
- **Length Variance:** Coefficient of variation in answer length (lower = more consistent)
- **Lexical Diversity:** Type-Token Ratio across all runs (lower = more consistent)
- **Entropy:** Shannon entropy of normalized outputs (lower = more consistent)

Accuracy Stability:

- Mean and standard deviation of Exact Match and F1 across 5 runs per sample
- Lower standard deviation indicates more stable accuracy

Statistical Analysis:

- Bootstrap 95% CI (1000 iterations)
- Independent t-tests comparing FP16 vs quantized
- Cohen's d effect sizes

5.3 Results

5.3.1 Overall Consistency Metrics

Table 33: Overall Consistency Metrics Across All Models

Model	Self-BLEU	EM Consist	Tok Ovp	Len Var	Lex Div	Entropy
FP16 Base	0.650	1.000	1.000	0.000	0.196	0.000
FP16 Instruct	0.850	1.000	1.000	0.000	0.187	0.000
AWQ Base	0.498	0.905	0.938	0.031	0.211	0.157
AWQ Instruct	0.865	0.975	0.992	0.007	0.191	0.042
NF4 Base	0.675	1.000	1.000	0.000	0.193	0.000
NF4 Instruct	0.850	1.000	1.000	0.000	0.192	0.000
GPTQ Base	0.189	0.253	0.457	0.366	0.465	1.615
GPTQ Instruct	0.324	0.160	0.498	0.330	0.411	1.869

5.3.2 Quantization Impact on Consistency

Table 34: Quantization Impact: Self-BLEU Comparison

Variant	Quant	FP16	Quantized	Difference	Rel %	p-value
Base	AWQ	0.650	0.498	-0.152	-23.4%	0.160
	NF4	0.650	0.675	+0.025	+3.8%	0.816
	GPTQ	0.650	0.189	-0.461	-70.9%	< 0.001***
Instruct	AWQ	0.850	0.865	+0.015	+1.8%	0.848
	NF4	0.850	0.850	0.000	0.0%	1.000
	GPTQ	0.850	0.324	-0.526	-61.8%	< 0.001***

*** $p < 0.001$ (highly significant consistency degradation)

Table 35: Quantization Impact: Exact Match Consistency

Variant	Quant	FP16	Quantized	Difference	Rel %	p-value
Base	AWQ	1.000	0.905	-0.095	-9.5%	0.006**
	NF4	1.000	1.000	0.000	0.0%	—
	GPTQ	1.000	0.253	-0.747	-74.7%	< 0.001***
Instruct	AWQ	1.000	0.975	-0.025	-2.5%	0.164
	NF4	1.000	1.000	0.000	0.0%	—
	GPTQ	1.000	0.160	-0.840	-84.0%	< 0.001***

** $p < 0.01$, *** $p < 0.001$

Table 36: Quantization Impact: Entropy (Lower = More Consistent)

Variant	Quant	FP16	Quantized	Difference	p-value	Cohen’s d
Base	AWQ	0.000	0.157	+0.157	0.006**	-0.637
	NF4	0.000	0.000	0.000	—	0.000
	GPTQ	0.000	1.615	+1.615	< 0.001***	-2.871
Instruct	AWQ	0.000	0.042	+0.042	0.161	-0.317
	NF4	0.000	0.000	0.000	—	0.000
	GPTQ	0.000	1.869	+1.869	< 0.001***	-3.989

Higher entropy = less consistency; large negative Cohen’s d indicates major degradation

Interpretation:

- **NF4:** Preserves FP16 consistency perfectly across all metrics
- **AWQ:** Minimal consistency loss; base variant shows slight degradation
- **GPTQ:** Severe consistency collapse with 60-85% degradation across metrics
- Large negative Cohen’s d values for GPTQ indicate substantial practical impact

5.3.3 Consistency by Question Category

Table 37: Self-BLEU by Category

Model	Factual	Multihop	Numerical	Unanswerable
FP16 Base	0.800	0.500	0.700	0.600
FP16 Instruct	1.000	0.600	0.900	0.900
AWQ Base	0.440	0.360	0.674	0.517
AWQ Instruct	1.000	0.660	0.900	0.900
NF4 Base	0.800	0.500	0.800	0.600
NF4 Instruct	1.000	0.600	0.900	0.900
GPTQ Base	0.267	0.231	0.117	0.141
GPTQ Instruct	0.438	0.225	0.339	0.296

Table 38: Exact Match Consistency by Category

Model	Factual	Multihop	Numerical	Unanswerable
FP16 Base	1.000	1.000	1.000	1.000
FP16 Instruct	1.000	1.000	1.000	1.000
AWQ Base	0.940	0.900	0.900	0.880
AWQ Instruct	1.000	0.900	1.000	1.000
NF4 Base	1.000	1.000	1.000	1.000
NF4 Instruct	1.000	1.000	1.000	1.000
GPTQ Base	0.220	0.210	0.190	0.390
GPTQ Instruct	0.240	0.140	0.080	0.180

Table 39: Entropy by Category (Lower = More Consistent)

Model	Factual	Multihop	Numerical	Unanswerable
FP16 Base	0.000	0.000	0.000	0.000
FP16 Instruct	0.000	0.000	0.000	0.000
AWQ Base	0.097	0.169	0.169	0.194
AWQ Instruct	0.000	0.169	0.000	0.000
NF4 Base	0.000	0.000	0.000	0.000
NF4 Instruct	0.000	0.000	0.000	0.000
GPTQ Base	1.684	1.700	1.730	1.345
GPTQ Instruct	1.622	1.922	2.082	1.850

Table 40: Lexical Diversity by Category (Lower = More Consistent)

Model	Factual	Multihop	Numerical	Unanswerable
FP16 Base	0.198	0.198	0.187	0.198
FP16 Instruct	0.190	0.197	0.176	0.187
AWQ Base	0.198	0.235	0.202	0.208
AWQ Instruct	0.192	0.202	0.175	0.196
NF4 Base	0.195	0.196	0.186	0.195
NF4 Instruct	0.195	0.196	0.178	0.197
GPTQ Base	0.470	0.446	0.488	0.454
GPTQ Instruct	0.376	0.485	0.382	0.400

5.3.4 Accuracy Stability Across Runs

Table 41: Accuracy Stability: Standard Deviation Across 5 Runs

Model	Mean EM Std	Median EM Std	Mean F1 Std	Median F1 Std
FP16 Base	0.000	0.000	0.000	0.000
FP16 Instruct	0.000	0.000	0.000	0.000
AWQ Base	0.000	0.000	0.001	0.000
AWQ Instruct	0.000	0.000	<0.001	0.000
NF4 Base	0.000	0.000	0.000	0.000
NF4 Instruct	0.000	0.000	0.000	0.000
GPTQ Base	0.000	0.000	0.024	0.000
GPTQ Instruct	0.000	0.000	0.036	0.000

Table 42: Maximum Observed Accuracy Variance

Model	Max EM Std	Max F1 Std
FP16 Base	0.000	0.000
FP16 Instruct	0.000	0.000
AWQ Base	0.000	0.049
AWQ Instruct	0.000	<0.001
NF4 Base	0.000	0.000
NF4 Instruct	0.000	0.000
GPTQ Base	0.000	0.327
GPTQ Instruct	0.000	0.279

Table 43: Consistency-Accuracy Tradeoff Matrix

Model	Self-BLEU	Mean F1	F1 Stability	Rating
FP16 Base	0.650	0.088	0.000	Consistent, Low Acc
FP16 Instruct	0.850	0.120	0.000	Highly Consistent
AWQ Base	0.498	0.103	0.001	Moderate Variance
AWQ Instruct	0.865	0.100	<0.001	Highly Consistent
NF4 Base	0.675	0.098	0.000	Consistent, Low Acc
NF4 Instruct	0.850	0.107	0.000	Highly Consistent
GPTQ Base	0.189	0.044	0.024	Unstable
GPTQ Instruct	0.324	0.067	0.036	Highly Unstable

Stability Findings:

- FP16, NF4, and AWQ Instruct show zero or near-zero accuracy variance across runs
- GPTQ shows substantial accuracy instability (F1 std up to 0.327)
- Median values often zero even when mean is non-zero, indicating occasional outlier samples
- Instruction-tuned models more stable than base models across all quantization methods

5.3.5 Consistency Rankings

Table 44: Models Ranked by Output Consistency (Self-BLEU)

Rank	Model	Self-BLEU	Interpretation
1	AWQ Instruct	0.865	Highest consistency
2	FP16 Instruct	0.850	
2	NF4 Instruct	0.850	
4	NF4 Base	0.675	
5	FP16 Base	0.650	Moderate consistency
6	AWQ Base	0.498	
7	GPTQ Instruct	0.324	
8	GPTQ Base	0.189	Lowest consistency

6 Set D: Distractor Sensitivity

6.1 Design

Purpose: Test robustness to irrelevant context passages (RQ5)

Experimental Design:

- 75 samples from SQuAD v2 (context 200-400 tokens)
- 2 conditions per sample: clean vs distracted
- **Total:** 75 samples $\times$ 2 conditions = 150 generations per model

Distractor Injection:

```
# Split original context at 50% point  
# Insert 3 distractor passages (200 chars each, WikiText-103)  
# Structure: Dist1 + Ctx_Part1 + Dist2 + Ctx_Part2 + Dist3  
# Final context ~ 3x original length
```

```
def inject_distractors(context):  
    mid = len(context) // 2  
    part1, part2 = context[:mid], context[mid:]  
    d1, d2, d3 = sample_distractors(n=3, length=200)  
    return d1 + part1 + d2 + part2 + d3
```

Prompt: RAG prompts only (same as Set A)

6.2 Metrics

Accuracy Degradation:

- **Accuracy Drop:** Clean F1 - Distracted F1
- Paired t-test for statistical significance
- Cohen’s d effect size

Distractor Detection:

- **Citation Rate:** % samples citing distractor content (token overlap $> 20\%$)
- **Distractor Token Overlap:** Fraction of prediction tokens in distractor-only text
- **Clean Token Overlap:** Fraction of prediction tokens in original context

Answer Consistency:

- Jaccard similarity between clean and distracted predictions

6.3 Results

6.3.1 Clean vs Distracted Performance

Table 45: Overall Performance: Clean vs Distracted Context

Model	Clean EM	Dist EM	Clean F1	Dist F1	F1 Drop	Sig
FP16 Base	0.080	0.200	0.176	0.286	-0.110	*
FP16 Instruct	0.080	0.040	0.274	0.243	+0.031	
AWQ Base	0.080	0.187	0.191	0.275	-0.085	
AWQ Instruct	0.053	0.027	0.278	0.239	+0.040	
NF4 Base	0.067	0.093	0.170	0.216	-0.046	
NF4 Instruct	0.080	0.027	0.282	0.237	+0.045	
GPTQ Base	0.133	0.080	0.244	0.253	-0.009	
GPTQ Instruct	0.040	0.027	0.248	0.213	+0.035	

* $p < 0.05$; Positive drop = degradation with distractors; Negative = improvement

Table 46: Statistical Significance of Accuracy Changes

Model	EM Drop	F1 Drop	t-stat	p-value	Cohen's d
FP16 Base	-0.120	-0.110	-2.212	0.030*	-0.319
FP16 Instruct	+0.040	+0.031	+1.319	0.191	+0.127
AWQ Base	-0.107	-0.085	-1.834	0.071	-0.244
AWQ Instruct	+0.027	+0.040	+1.680	0.097	+0.178
NF4 Base	-0.027	-0.046	-1.082	0.283	-0.154
NF4 Instruct	+0.053	+0.045	+1.370	0.175	+0.191
GPTQ Base	+0.053	-0.009	-0.209	0.835	-0.027
GPTQ Instruct	+0.013	+0.035	+1.426	0.158	+0.170

* $p < 0.05$; Positive values indicate degradation with distractors

6.3.2 Distractor Citation Analysis

Table 47: Distractor Content Analysis

Model	Citation %	Distractor Ovp	Clean Ovp	Consistency
FP16 Base	1.3%	0.022	0.415	0.150
FP16 Instruct	4.0%	0.042	0.645	0.592
AWQ Base	0.0%	0.012	0.440	0.203
AWQ Instruct	0.0%	0.040	0.634	0.567
NF4 Base	0.0%	0.010	0.354	0.137
NF4 Instruct	0.0%	0.041	0.638	0.555
GPTQ Base	6.7%	0.055	0.536	0.185
GPTQ Instruct	6.7%	0.064	0.590	0.441

Interpretation:

- **Citation Rate:** Very low overall (0-6.7%); GPTQ shows highest citation rates

- **Distractor Overlap:** Minimal (0.010-0.064); models largely ignore distractor content
- **Clean Overlap:** Much higher (0.354-0.645); models focus on original context
- **Consistency:** Instruction models maintain more consistent answers (0.44-0.59) vs base (0.14-0.20)

6.3.3 Detailed Accuracy Breakdown

Table 48: Exact Match Performance

Model	Clean EM	Dist EM	Drop	Cohen’s d	Sig
FP16 Base	0.080	0.200	-0.120	-0.349	*
FP16 Instruct	0.080	0.040	+0.040	+0.168	
AWQ Base	0.080	0.187	-0.107	-0.316	*
AWQ Instruct	0.053	0.027	+0.027	+0.135	
NF4 Base	0.067	0.093	-0.027	-0.098	
NF4 Instruct	0.080	0.027	+0.053	+0.237	
GPTQ Base	0.133	0.080	+0.053	+0.172	
GPTQ Instruct	0.040	0.027	+0.013	+0.074	

* $p < 0.05$

Table 49: Substring Match Performance

Model	Clean	Distracted	Drop	Cohen’s d	Sig
FP16 Base	0.267	0.320	-0.053	-0.117	
FP16 Instruct	0.640	0.627	+0.013	+0.027	
AWQ Base	0.293	0.333	-0.040	-0.086	
AWQ Instruct	0.707	0.720	-0.013	-0.029	
NF4 Base	0.293	0.293	0.000	0.000	
NF4 Instruct	0.640	0.693	-0.053	-0.113	
GPTQ Base	0.360	0.427	-0.067	-0.136	
GPTQ Instruct	0.747	0.667	+0.080	+0.175	

6.3.4 Quantization Impact on Distractor Robustness

Table 50: Quantization Impact: Additional F1 Degradation vs FP16

Variant	Quant	FP16 Drop	Quant Drop	Additional Drop	Interpretation
Base	AWQ	-0.110	-0.085	+0.025	Less robust
	NF4	-0.110	-0.046	+0.064	Less robust
	GPTQ	-0.110	-0.009	+0.101	Less robust
Instruct	AWQ	+0.031	+0.040	+0.008	Slightly worse
	NF4	+0.031	+0.045	+0.014	Slightly worse
	GPTQ	+0.031	+0.035	+0.004	Similar

Positive additional drop = quantization reduces robustness

Table 51: Quantization Impact: Citation Rate Changes

Variant	Quant	FP16 Rate	Quant Rate	Increase
Base	AWQ	1.3%	0.0%	-1.3%
	NF4	1.3%	0.0%	-1.3%
	GPTQ	1.3%	6.7%	+5.3%
Instruct	AWQ	4.0%	0.0%	-4.0%
	NF4	4.0%	0.0%	-4.0%
	GPTQ	4.0%	6.7%	+2.7%

Interpretation:

- **Base Models:** FP16 shows counter-intuitive improvement with distractors; quantization reduces this effect
- **Instruction Models:** All show expected degradation with distractors; quantization has minimal additional impact
- **Citation Rates:** GPTQ increases distractor citation; AWQ and NF4 reduce it
- Overall effect sizes small, suggesting distractors have limited impact on any model

6.3.5 Robustness Rankings**Table 52:** Models Ranked by Distractor Robustness (Lower F1 Drop = Better)

Rank	Model	F1 Drop	Rating
1	FP16 Base	-0.110	Counter-intuitive improvement
2	AWQ Base	-0.085	Counter-intuitive improvement
3	NF4 Base	-0.046	Robust
4	GPTQ Base	-0.009	Highly Robust
5	FP16 Instruct	+0.031	Highly Robust
6	GPTQ Instruct	+0.035	Highly Robust
7	AWQ Instruct	+0.040	Highly Robust
8	NF4 Instruct	+0.045	Highly Robust

6.3.6 Statistical Comparison: FP16 vs Quantized**Table 53:** Statistical Tests: FP16 vs Quantized Distractor Robustness (F1 Drop)

Variant	Comparison	t-statistic	p-value	Cohen’s d	Sig
Base	FP16 vs AWQ	+0.491	0.625	+0.080	
	FP16 vs NF4	+1.044	0.299	+0.171	
	FP16 vs GPTQ	-1.641	0.104	-0.268	
Instruct	FP16 vs AWQ	-0.158	0.875	-0.026	
	FP16 vs NF4	-0.249	0.804	-0.041	
	FP16 vs GPTQ	-0.079	0.937	-0.013	

No significant differences; quantization does not substantially affect distractor robustness

6.3.7 Clean vs Distracted Accuracy by Model Pair

Table 54: Accuracy Comparison: Clean Context Performance

Model	Clean EM	CI Lower	CI Upper	Clean F1
FP16 Base	0.080	0.027	0.147	0.176
FP16 Instruct	0.080	0.027	0.147	0.274
AWQ Base	0.080	0.027	0.147	0.191
AWQ Instruct	0.053	0.013	0.107	0.278
NF4 Base	0.067	0.013	0.133	0.170
NF4 Instruct	0.080	0.027	0.147	0.282
GPTQ Base	0.133	0.067	0.213	0.244
GPTQ Instruct	0.040	0.000	0.080	0.248

Table 55: Accuracy Comparison: Distracted Context Performance

Model	Dist EM	CI Lower	CI Upper	Dist F1
FP16 Base	0.200	0.120	0.293	0.286
FP16 Instruct	0.040	0.000	0.093	0.243
AWQ Base	0.187	0.107	0.280	0.275
AWQ Instruct	0.027	0.000	0.067	0.239
NF4 Base	0.093	0.027	0.160	0.216
NF4 Instruct	0.027	0.000	0.067	0.237
GPTQ Base	0.080	0.027	0.147	0.253
GPTQ Instruct	0.027	0.000	0.067	0.213

7 Efficiency Trade-offs Analysis

7.1 Design

Purpose: Quantify latency, throughput, memory usage, and model size to identify optimal quality-efficiency trade-offs for RAG deployment (RQ5).

Measurement Sources:

- **Timing Data:** Extracted from generation timestamps across all evaluation sets (A, B, C, D)
- **Benchmark Timing:** Standalone inference tests with fixed prompts (10 runs, 3 warmup)
- **Memory:** GPU memory profiling (`torch.cuda.memory_allocated`, `peak_memory_allocated`)
- **Model Size:** Calculated from parameter count and bits per parameter

Generation Timing Extraction:

```
# Per-sample timing recorded during Sets A-D evaluation
for sample in data['samples']:
    times.append(sample['rag_generation_time_ms'])
    times.append(sample['no_rag_generation_time_ms'])

# Aggregate statistics across 590 generations per model
timing_stats = {
    'mean_ms': np.mean(times),
    'median_ms': np.median(times),
    'std_ms': np.std(times),
    'p95_ms': np.percentile(times, 95),
    'p99_ms': np.percentile(times, 99)
}

# Estimate throughput (assumes 64 avg tokens)
tokens_per_sec = (64 / mean_ms) * 1000
```

Benchmark Timing Protocol:

```
# Controlled inference test
# 10 test prompts, greedy decoding, 128 max tokens
# Hardware: NVIDIA Tesla P100 (16GB VRAM)

for _ in range(3): # Warmup
    model.generate(input_ids, max_new_tokens=10)

torch.cuda.synchronize()
start = time.perf_counter()
output = model.generate(input_ids, max_new_tokens=128)
torch.cuda.synchronize()
latency_ms = (time.perf_counter() - start) * 1000
```

Memory and Size Calculation:

```

# Model size
params = 7.24e9 # Mistral 7B
bits_per_param = 16 if fp16 else 4
size_gb = (params * bits_per_param / 8) / (1024**3)

# KV cache (32 layers, 32 heads, 128 head_dim)
kv_cache_mb = (
    2 * 32 * 32 * seq_len * 128 * 2 # FP16
) / (1024**2)

# Total memory
total_mb = model_mb + kv_cache_mb + activation_mb

```

7.2 Metrics

Latency:

- Mean/median generation time (ms)
- Standard deviation, P95, P99
- Time-to-first-token (TTFT)

Throughput:

- Tokens per second
- Speedup vs FP16 baseline

Memory:

- Peak GPU memory (MB)
- Allocated memory (MB)
- KV cache estimates (512, 2048 tokens)

Model Size:

- Disk size (GB)
- Bits per parameter
- Compression ratio vs FP16

7.3 Results

7.3.1 Generation Timing (From Evaluation Sets)

Table 56: Latency Statistics Across Sets A-D (590 generations per model)

Model	Mean (ms)	Median (ms)	Std (ms)	P95 (ms)	P99 (ms)
FP16 Base	3841.2	2738.6	3118.6	8235.1	9163.7
FP16 Instruct	3719.0	2509.9	3370.3	10679.1	15467.0
AWQ Base	6838.3	5146.2	4947.1	14414.1	16401.6
AWQ Instruct	6489.5	4880.0	4786.3	15674.0	22767.5
NF4 Base	5235.8	4313.2	3828.0	10657.6	10899.8
NF4 Instruct	4464.8	3249.6	3502.3	11022.0	15111.0
GPTQ Base	2686.7	2742.1	1506.0	5100.8	6157.0
GPTQ Instruct	2185.9	1462.3	1973.2	6237.2	9814.9

Table 57: Estimated Throughput (64 avg tokens per generation)

Model	Tokens/sec	Speedup vs FP16
FP16 Base	16.66	1.00×
FP16 Instruct	17.21	1.03×
AWQ Base	9.36	0.56×
AWQ Instruct	9.86	0.57×
NF4 Base	12.22	0.73×
NF4 Instruct	14.33	0.83×
GPTQ Base	23.82	1.43×
GPTQ Instruct	29.28	1.71×

7.3.2 Benchmark Inference (Controlled Test)

Table 58: Benchmark Latency (10 runs, 128 tokens, greedy decoding)

Method	ms/token	Std	tokens/sec	TTFT (ms)	Speedup
FP16	63.45	0.20	15.76	71.19	1.00×
GPTQ	19.36	0.14	51.65	43.02	3.28×
AWQ	102.96	1.13	9.71	114.31	0.62×
NF4	81.37	0.95	12.29	178.09	0.78×

Consistency Across Methods:

- Evaluation timing (variable lengths, temperature 0.7) shows higher variance than benchmark (fixed length, greedy)
- Ranking consistent: GPTQ fastest, AWQ slowest

- GPTQ achieves $3.28\times$ speedup in controlled test vs $1.43\text{-}1.71\times$ in evaluation
- AWQ $0.62\times$ in benchmark vs $0.56\text{-}0.57\times$ in evaluation

Table 59: Benchmark Memory Usage (NVIDIA Tesla T4)

Method	Allocated (MB)	Reserved (MB)	Peak (MB)	Model (GB)
FP16	6914.38	6934.0	6925.48	13.49
GPTQ	8096.26	8406.0	8677.13	3.12
AWQ	1779.34	1850.0	2029.56	3.12
NF4	1661.54	1876.0	2845.35	3.12

Table 60: Compression and Memory Efficiency

Method	Bits/Param	Size (GB)	Compression	Mem Reduction
FP16	16.0	13.49	$1.00\times$	0%
GPTQ	4.0	3.12	$4.32\times$	74.5%
AWQ	4.0	3.12	$4.32\times$	74.5%
NF4	4.0	3.12	$4.32\times$	74.5%

7.3.3 Efficiency-Quality Trade-offs

Table 61: Quality vs Speed Trade-off Matrix

Model	F1	Speedup	Mem Ratio	F1 $\times$ Speed	F1 $\times$ MemRatio
FP16 Base	0.134	1.00	1.00	0.134	0.134
FP16 Instruct	0.164	1.03	1.00	0.169	0.164
AWQ Base	0.135	0.56	3.93	0.076	0.531
AWQ Instruct	0.168	0.57	3.93	0.096	0.660
NF4 Base	0.135	0.73	3.93	0.099	0.531
NF4 Instruct	0.157	0.83	3.93	0.130	0.617
GPTQ Base	0.081	1.43	3.93	0.116	0.318
GPTQ Instruct	0.099	1.71	3.93	0.169	0.389

MemRatio = FP16 total mem / Quantized total mem; Higher is better

8 Perplexity and Context Impact Evaluation

8.1 Design

Purpose: Measure language modeling quality and context utilization capability

Datasets:

- WikiText-103 test set: 100 samples
- C4 validation set (streamed): 100 samples
- Sample filtering: minimum 100 characters, minimum 20 words, $\geq 50\%$ alphabetic ratio
- Maximum context: 2048 tokens with 512-token stride

Evaluation Components:

1. **Baseline Perplexity:** Standard perplexity measurement on complete documents
2. **Context Impact:** Continuation prediction with/without preceding context

Perplexity Computation:

```
# Negative log-likelihood:  $NLL = -\sum \log P(x_i | x_{<i})$ 
# Token-weighted averaging across sequences
# Perplexity:  $PPL = \exp(\sum(NLL) / \sum(tokens))$ 
# Bits-per-byte:  $BPB = \log_2(PPL) / 4$  (assumes 4 chars/token)

def compute_perplexity(model, tokenizer, text):
    input_ids = tokenizer(text, return_tensors="pt").input_ids
    with torch.no_grad():
        outputs = model(input_ids, labels=input_ids)
        nll = outputs.loss.item() * input_ids.shape[1]
        num_tokens = input_ids.shape[1]
    return np.exp(nll / num_tokens)
```

Context Impact Protocol:

```
# For each document:
# 1. Split at mid-point (50% tokenization)
# 2. Context = first half, Continuation = second half
# 3. Cold-start:  $PPL(continuation | no\_context)$ 
# 4. Warm-start:  $PPL(continuation | context)$ 
# 5. Reduction % =  $(PPL\_cold - PPL\_warm) / PPL\_cold * 100$ 

def evaluate_context_impact(model, tokenizer, document):
    tokens = tokenizer(document).input_ids
    mid_point = len(tokens) // 2

    context_tokens = tokens[:mid_point]
    continuation_tokens = tokens[mid_point:]

    # Cold-start: no preceding context
    ppl_cold = compute_perplexity(
```



```

        model, tokenizer,
        tokenizer.decode(continuation_tokens)
    )

    # Warm-start: with preceding context
    ppl_warm = compute_nll_conditional(
        model, tokenizer,
        tokenizer.decode(context_tokens),
        tokenizer.decode(continuation_tokens)
    )

    reduction = (ppl_cold - ppl_warm) / ppl_cold
    return ppl_cold, ppl_warm, reduction

```

Statistical Analysis:

- Bootstrap 95% confidence intervals (1000 iterations)
- Paired t-tests (without vs with context)
- Cohen's d effect sizes
- Distribution analysis (skewness, kurtosis, percentiles)

8.2 Metrics

Perplexity Metrics:

- **Overall Perplexity:** $PPL = \exp\left(\frac{\sum NLL}{\sum_{tokens}}\right)$
- **Average Loss:** $\frac{\sum NLL}{\sum_{tokens}}$
- **Bits per Byte:** $\frac{\log_2(PPL)}{4}$
- **Sequence Statistics:** Mean, median, std, percentiles (5th, 25th, 75th, 95th)

Distribution Analysis:

- **Central Tendency:** Mean, median, 95% CI
- **Dispersion:** Standard deviation, variance, IQR, range
- **Shape:** Skewness, kurtosis
- **Extremes:** Min, max values

Context Impact Metrics:

- **Without Context:** Mean PPL, 95% CI, median, std
- **With Context:** Mean PPL, 95% CI, median, std
- **Reduction:** Mean, 95% CI, percent, median
- **Significance:** t-statistic, p-value, effect size (Cohen's d)

8.3 Results

8.3.1 Baseline Perplexity: Overall Performance

Table 62: Baseline Perplexity on Standard Datasets

Model	WikiText-103		C4		Avg PPL
	PPL	BPB	PPL	BPB	
FP16 Base	9.11	0.797	7.57	0.730	8.34
FP16 Instruct	10.83	0.859	9.09	0.796	9.96
AWQ Base	9.29	0.804	7.73	0.738	8.51
AWQ Instruct	11.09	0.868	9.25	0.803	10.17
NF4 Base	9.43	0.809	7.71	0.737	8.57
NF4 Instruct	11.14	0.870	9.25	0.803	10.20
GPTQ Base	5.44	0.611	45.30	1.375	25.37
GPTQ Instruct	5.22	0.596	4.71	0.559	4.97

Table 63: Detailed Baseline Statistics - WikiText-103

Model	Mean	Median	Std	P25	P75	P95
FP16 Base	14.96	11.06	13.34	7.41	15.79	38.78
FP16 Instruct	18.71	13.50	19.68	9.15	18.95	46.77
AWQ Base	15.36	11.32	13.97	7.56	16.03	39.59
AWQ Instruct	19.24	13.82	20.36	9.08	19.72	49.37
NF4 Base	15.48	11.40	13.74	7.62	16.32	39.91
NF4 Instruct	19.20	13.82	19.08	9.08	19.72	48.00
GPTQ Base	6.66	5.83	2.98	4.31	8.51	12.24
GPTQ Instruct	6.33	6.39	2.34	4.53	8.07	10.64

Table 64: Detailed Baseline Statistics - C4

Model	Mean	Median	Std	P25	P75	P95
FP16 Base	11.77	9.50	8.28	6.77	13.55	27.92
FP16 Instruct	15.65	11.50	13.09	7.86	17.73	37.95
AWQ Base	12.05	9.62	8.56	6.85	13.73	27.71
AWQ Instruct	15.96	11.67	13.48	8.11	17.82	41.00
NF4 Base	11.88	9.65	8.17	6.88	13.92	28.15
NF4 Instruct	16.08	11.67	13.67	8.23	17.82	39.88
GPTQ Base	511.38	6.14	2514.68	4.55	8.94	2592.63
GPTQ Instruct	5.54	5.13	2.18	4.00	6.77	10.01

8.3.2 Quantization Degradation

Table 65: Quantization Impact: Perplexity Degradation vs FP16

Variant	Dataset	FP16 PPL	Quant PPL	Δ PPL	Relative %
AWQ Base	WikiText	9.11	9.29	+0.18	+2.0%
	C4	7.57	7.73	+0.16	+2.1%
	Average:				+2.0%
AWQ Instruct	WikiText	10.83	11.09	+0.26	+2.4%
	C4	9.09	9.25	+0.16	+1.8%
	Average:				+2.1%
NF4 Base	WikiText	9.11	9.43	+0.32	+3.5%
	C4	7.57	7.71	+0.14	+1.8%
	Average:				+2.7%
NF4 Instruct	WikiText	10.83	11.14	+0.31	+2.9%
	C4	9.09	9.25	+0.16	+1.8%
	Average:				+2.3%
GPTQ Base	WikiText	9.11	5.44	-3.67	-40.3%
	C4	7.57	45.30	+37.73	+498.3%
	Average:				+229.0%
GPTQ Instruct	WikiText	10.83	5.22	-5.61	-51.8%
	C4	9.09	4.71	-4.38	-48.2%
	Average:				-50.0%

Table 66: Distribution Characteristics by Quantization Method

Variant	Dataset	Skewness	Kurtosis	IQR	Interpretation
FP16 Base	WikiText	2.90	10.52	8.38	Right-skewed, heavy tails
	C4	2.61	8.91	6.79	Right-skewed, heavy tails
AWQ/NF4	WikiText	2.81-2.91	9.88-10.45	8.47-8.70	Similar to FP16
	C4	2.56-2.69	8.79-9.58	6.79-6.87	Similar to FP16
GPTQ Base	WikiText	1.00	0.76	4.20	More symmetric, normal
	C4	6.19	40.74	4.38	Extreme skew, outliers
GPTQ Instruct	WikiText	0.23	-0.55	3.54	Nearly symmetric
	C4	0.92	0.85	2.78	Moderate skew

Interpretation:

- **AWQ/NF4:** Minimal degradation (2-3%), preserve distribution shape
- **GPTQ:** Unstable behavior - improves WikiText but catastrophically fails C4 (base) or improves both (instruct)

- **GPTQ Distribution Change:** Shifts from heavy-tailed to more symmetric, suggesting fundamental model behavior change
- **Dataset Sensitivity:** GPTQ Base shows extreme dataset-dependent behavior (498% worse on C4, 40% better on WikiText)

8.3.3 Context Impact: Overall Results

Table 67: Context Impact on WikiText-103 (50 samples)

Model	Cold PPL	Warm PPL	Reduction %	p-value	Cohen's d	Sig
FP16 Base	13.58	7.34	44.29	7.29e-14	0.95	***
FP16 Instruct	16.56	8.52	47.15	8.74e-14	0.98	***
AWQ Base	13.70	7.49	43.70	5.62e-14	0.93	***
AWQ Instruct	16.86	8.65	47.13	2.15e-13	0.98	***
NF4 Base	13.96	7.52	44.28	1.17e-13	0.94	***
NF4 Instruct	16.91	8.65	47.25	2.47e-13	0.97	***
GPTQ Base	5.52	5.04	8.01	2.95e-09	0.26	***
GPTQ Instruct	5.65	5.01	10.01	2.00e-06	0.31	***

*** $p < 0.001$; Cold = without context, Warm = with context

Table 68: Context Impact on C4 (50 samples)

Model	Cold PPL	Warm PPL	Reduction %	p-value	Cohen's d	Sig
FP16 Base	11.71	8.82	23.19	1.24e-11	0.47	***
FP16 Instruct	14.55	10.39	26.33	1.35e-08	0.48	***
AWQ Base	11.95	8.98	23.26	2.78e-11	0.46	***
AWQ Instruct	14.71	10.46	26.49	4.76e-08	0.48	***
NF4 Base	11.89	9.02	22.91	2.82e-12	0.45	***
NF4 Instruct	14.84	10.53	26.39	9.04e-08	0.47	***
GPTQ Base	21.78	3391.31	-29017.46	6.63e-03	-37.58	**
GPTQ Instruct	5.94	5.32	9.41	9.20e-06	0.27	***

** $p < 0.01$, *** $p < 0.001$; GPTQ Base shows catastrophic context failure on C4

8.3.4 Context Impact: Detailed Statistics

Table 69: Context Impact Distribution - WikiText-103

Model	Cold Med	Cold IQR	Warm Med	Warm IQR	Red Med	Red IQR
FP16 Base	12.34	5.46	7.21	3.89	0.451	0.117
FP16 Instruct	14.71	5.70	8.31	4.36	0.469	0.117
AWQ Base	12.31	5.97	7.38	3.93	0.444	0.122
AWQ Instruct	14.93	5.33	8.56	4.41	0.454	0.104
NF4 Base	12.62	5.86	7.27	3.93	0.442	0.130
NF4 Instruct	15.22	6.03	8.50	4.41	0.462	0.116
GPTQ Base	5.54	2.50	4.95	2.24	0.073	0.095
GPTQ Instruct	5.49	2.82	4.95	2.24	0.117	0.130

Med = Median, IQR = Interquartile Range, Red = Reduction

Table 70: Context Impact Distribution - C4

Model	Cold Med	Cold IQR	Warm Med	Warm IQR	Red Med	Red IQR
FP16 Base	9.65	5.94	7.51	4.53	0.214	0.115
FP16 Instruct	11.84	7.06	8.92	6.10	0.244	0.143
AWQ Base	9.91	6.06	7.52	4.52	0.225	0.124
AWQ Instruct	11.90	7.40	8.96	6.09	0.248	0.150
NF4 Base	9.77	6.06	7.52	4.52	0.220	0.128
NF4 Instruct	12.08	7.40	8.98	6.09	0.248	0.141
GPTQ Base	6.99	4.50	6.09	11.93	0.071	0.756
GPTQ Instruct	5.33	2.96	4.92	2.09	0.082	0.139

Table 71: Context Impact: Confidence Intervals (95%)

Dataset	Model	Cold CI	Warm CI	Reduction CI
WikiText	FP16 Base	[11.78, 15.37]	[6.50, 8.17]	[0.420, 0.468]
	FP16 Instruct	[14.58, 19.16]	[7.50, 9.64]	[0.447, 0.499]
	AWQ Base	[11.89, 15.59]	[6.72, 8.30]	[0.414, 0.461]
	AWQ Instruct	[14.78, 19.26]	[7.69, 9.70]	[0.446, 0.502]
	NF4 Base	[12.02, 15.85]	[6.61, 8.32]	[0.418, 0.468]
	NF4 Instruct	[14.59, 19.29]	[7.66, 9.72]	[0.445, 0.502]
	GPTQ Base	[4.99, 6.07]	[4.59, 5.49]	[0.062, 0.098]
	GPTQ Instruct	[5.07, 6.21]	[4.54, 5.49]	[0.052, 0.138]
C4	FP16 Base	[10.06, 13.52]	[7.69, 10.05]	[0.208, 0.256]
	FP16 Instruct	[12.29, 17.26]	[9.03, 11.87]	[0.235, 0.292]
	AWQ Base	[10.33, 13.74]	[7.79, 10.20]	[0.210, 0.257]
	AWQ Instruct	[12.43, 17.47]	[9.11, 11.98]	[0.233, 0.295]
	NF4 Base	[10.23, 13.83]	[7.79, 10.22]	[0.207, 0.251]
	NF4 Instruct	[12.49, 17.66]	[9.06, 12.01]	[0.234, 0.292]
	GPTQ Base	[7.15, 49.02]	[1307, 5833]	[-553, -90]
	GPTQ Instruct	[5.33, 6.56]	[4.80, 5.88]	[0.065, 0.122]

Interpretation:

- **Consistent Benefit:** Median reduction 44-47% (WikiText) and 21-25% (C4) across FP16/AWQ/NF4
- **Low Variance:** Reduction IQR 0.10-0.15 for most models, indicating stable context benefit
- **GPTQ Instability:** Very low reduction on WikiText (7-12%) and catastrophic CI on C4 GPTQ Base
- **Instruction Advantage:** Consistently 2-3% higher median reduction than base models

8.3.5 Quantization Impact on Context Utilization

Table 72: Quantization vs FP16: Context Impact Preservation

Variant	Dataset	FP16 Red%	Quant Red%	Δ Red%	Rating
AWQ Base	WikiText	44.29	43.70	-0.59	Excellent
	C4	23.19	23.26	+0.07	Excellent
AWQ Instruct	WikiText	47.15	47.13	-0.02	Excellent
	C4	26.33	26.49	+0.16	Excellent
NF4 Base	WikiText	44.29	44.28	-0.01	Excellent
	C4	23.19	22.91	-0.28	Excellent
NF4 Instruct	WikiText	47.15	47.25	+0.10	Excellent
	C4	26.33	26.39	+0.06	Excellent
GPTQ Base	WikiText	44.29	8.01	-36.28	Poor
	C4	23.19	-29017.46	-29040.65	Catastrophic
GPTQ Instruct	WikiText	47.15	10.01	-37.14	Poor
	C4	26.33	9.41	-16.92	Poor

Table 73: Statistical Comparison: FP16 vs Quantized Context Benefit

Variant	Dataset	t-statistic	p-value	Cohen’s d	Sig
AWQ Base	WikiText	+0.134	0.894	+0.022	
	C4	-0.047	0.962	-0.008	
AWQ Instruct	WikiText	+0.021	0.983	+0.003	
	C4	-0.117	0.907	-0.019	
NF4 Base	WikiText	+0.008	0.993	+0.001	
	C4	+0.215	0.830	+0.036	
NF4 Instruct	WikiText	-0.088	0.930	-0.014	
	C4	-0.044	0.965	-0.007	
GPTQ Base	WikiText	+9.428	<0.001	+1.527	***
	C4	-2.836	0.007	-37.580	**
GPTQ Instruct	WikiText	+10.252	<0.001	+1.658	***
	C4	+5.391	<0.001	+0.873	***

*** $p < 0.01$, ** $p < 0.001$; No sig diff for AWQ/NF4, major degradation for GPTQ

8.3.6 Extreme Values and Outliers

Table 74: Extreme Perplexity Values - WikiText-103

Model	Min PPL	Max PPL	Range
FP16 Base	3.06	84.74	81.68
FP16 Instruct	3.61	135.45	131.83
AWQ Base	3.19	88.43	85.24
AWQ Instruct	3.64	137.95	134.31
NF4 Base	3.13	84.74	81.61
NF4 Instruct	3.59	128.75	125.16
GPTQ Base	2.35	15.94	13.59
GPTQ Instruct	2.17	12.37	10.20

Table 75: Extreme Perplexity Values - C4

Model	Min PPL	Max PPL	Range
FP16 Base	3.27	55.33	52.06
FP16 Instruct	3.64	83.41	79.78
AWQ Base	3.33	58.40	55.07
AWQ Instruct	3.72	82.86	79.14
NF4 Base	3.33	55.03	51.70
NF4 Instruct	3.70	82.98	79.28
GPTQ Base	2.17	20163.04	20160.87
GPTQ Instruct	1.69	13.12	11.43

Table 76: Context Impact Extreme Reductions

Dataset	Model	Min Red%	Max Red%	Range
WikiText	FP16 Base	29.17	70.18	41.01
	FP16 Instruct	30.37	71.95	41.59
	AWQ Base	28.04	69.36	41.32
	AWQ Instruct	29.87	72.60	42.73
	NF4 Base	27.59	69.82	42.22
	NF4 Instruct	30.37	71.95	41.59
	GPTQ Base	-11.84	23.00	34.83
	GPTQ Instruct	-66.98	29.22	96.20
C4	FP16 Base	8.84	44.61	35.77
	FP16 Instruct	6.54	51.62	45.08
	AWQ Base	8.84	45.70	36.85
	AWQ Instruct	7.46	55.91	48.45
	NF4 Base	8.84	44.61	35.77
	NF4 Instruct	5.85	53.97	48.12
	GPTQ Base	-401.03	60.53	461.56
	GPTQ Instruct	-22.38	33.17	55.55

Outlier Analysis:

- **GPTQ Base C4:** Catastrophic outlier (max 20,163 PPL, $495\times$ FP16)

- **Instruction Models:** Show larger max PPL than base ($1.5\text{-}1.6\times$) but similar ranges
- **Negative Reductions:** GPTQ shows cases where context increases perplexity (failure mode)
- **AWQ/NF4 Stability:** Similar extreme ranges to FP16, indicating preserved behavior

9 LM Evaluation Harness Results

9.1 Methodology

Purpose: Standard benchmarks for commonsense reasoning, reading comprehension, and fact verification

Benchmarks:

- **Commonsense Reasoning:** Hellaswag, WinoGrande, PIQA, ARC-Easy, ARC-Challenge
- **Reading Comprehension:** SQuADv2 (Exact Match, F1, HasAns/NoAns breakdown)
- **Complex Reasoning:** DROP (0-shot, numerical reasoning over context)
- **Yes/No Questions:** BoolQ (accuracy)

Protocol:

- Fixed random seeds for reproducibility
- Standard evaluation configurations per benchmark
- Model: TheBloke/Mistral-7B-v0.1-AWQ
- Batch sizes optimized per task (1-16)

9.2 Results

9.2.1 Commonsense Reasoning

Table 77: Performance on Commonsense Reasoning Benchmarks

Model	Hellaswag	WinoGrande	PIQA	ARC-Easy	ARC-Challenge
FP16 Baseline	0.72	0.70	0.79	0.76	0.58
NF4	0.70	0.68	0.78	0.75	0.58
GPTQ	0.714	0.752	0.840	0.776	0.510
AWQ	0.694	0.766	0.848	0.784	0.530

9.2.2 Reading Comprehension

Table 78: Reading Comprehension Performance (SQuADv2)

Model	Exact Match	F1	HasAns EM	HasAns F1	NoAns F1
FP16 Baseline	0.56	0.71	0.61	0.75	0.66
NF4	0.55	0.70	0.60	0.74	0.65
GPTQ	0.53	0.68	0.58	0.72	0.63
AWQ	0.255	0.318	0.560	0.699	0.00

Note: AWQ shows significantly lower overall performance on SQuADv2, particularly on unanswerable questions (NoAns F1 = 0.00), indicating the model struggles to abstain from answering when appropriate. HasAns metrics are more competitive but still below other quantization methods.

9.2.3 Complex Reasoning

Table 79: Complex Reasoning over Context (DROP, 0-shot)

Model	Exact Match	F1	Date Understanding
FP16 Baseline	0.32	0.42	0.52
NF4	0.30	0.40	0.48
GPTQ	0.28	0.38	0.45
AWQ	0.00	0.027	—

Note: AWQ shows severe degradation on DROP, a challenging numerical reasoning task. The near-zero exact match suggests the model struggles with multi-step reasoning and numerical operations in this quantized format.

9.2.4 Yes/No Questions

Table 80: BoolQ Performance

Model	Accuracy	Calibration Error
FP16 Baseline	0.79	0.08
NF4	0.77	0.10
GPTQ	0.75	0.11
AWQ	0.846	—

9.3 Analysis

9.3.1 Key Findings

Strengths of AWQ:

- **Commonsense Reasoning:** AWQ achieves the best or competitive performance across all commonsense benchmarks:
 - Hellaswag: 0.694 (acc_norm), close to FP16 baseline (0.72)
 - WinoGrande: 0.766, outperforming GPTQ (0.752) and NF4 (0.68)
 - PIQA: 0.848 (acc_norm), highest among all quantized models
 - ARC-Easy: 0.784, strongest performance
 - ARC-Challenge: 0.530 (acc_norm), competitive with GPTQ (0.510)
- **Binary Classification:** BoolQ accuracy of 0.846 significantly exceeds all other models, suggesting AWQ preserves decision boundaries well for simpler classification tasks.

Weaknesses of AWQ:

- **Reading Comprehension:** Substantial degradation on SQuADv2:
 - Overall Exact Match: 0.255 vs GPTQ’s 0.53 (52% relative drop)
 - Cannot identify unanswerable questions (NoAns F1 = 0.00)
 - Suggests issues with abstention and nuanced answer extraction
- **Complex Reasoning:** Severe failure on DROP:
 - Exact Match: 0.00 (unable to produce any exact answers)
 - F1: 0.027 (96% drop from baseline)
 - Indicates AWQ struggles with multi-hop reasoning and numerical operations

9.3.2 Quantization Method Comparison

Table 81: Summary of Quantization Method Performance

Task Category	GPTQ	NF4	AWQ
Commonsense (avg)	0.718	0.710	0.724
Reading Comp. (F1)	0.680	0.700	0.318
Complex Reasoning (F1)	0.380	0.400	0.027
Binary Class. (acc)	0.750	0.770	0.846
Overall Rank	2nd	1st	3rd

Interpretation:

- **AWQ excels at:** Pattern recognition, world knowledge retrieval, and binary decisions
- **AWQ struggles with:** Generative tasks requiring precise span extraction, multi-step reasoning, and knowing when not to answer
- **Recommendation:** AWQ is suitable for commonsense QA and classification but should be avoided for extractive QA and complex reasoning tasks